

Daniil Parshin

Software Engineer | Systems & Performance

Professional Summary

C++ Software Engineer focused on systems and performance engineering on Linux. I develop and optimize performance-sensitive C/C++, Python, and Go software for profiling, graphics, and CPU inference, with experience in multithreading, benchmarking, build systems, CI/CD, debugging, and automated testing.

Experience

Huawei — Nizhny Novgorod, Russia • 2 yrs 4 mos

Software Engineer — Profiling Tools Team | Nov 2025 — Oct 2026

- Refactored the CMake superbuild of a performance profiling product comprising dozens of components, improving build speed by **~30%** and decoupling package assembly from component builds.
- Maintain C/C++, Python, and Go product code and Jenkins pipelines; automate builds, artifact handling, packaging, and Docker-based build/test environments.
- Develop unit tests with GoogleTest and integration tests with Robot Framework; automate test execution and reporting in CI.

Software Engineer — GPU Team | Jul 2024 — Nov 2025

- Improved BC1 texture compression quality by **~3%** through dithering techniques and integrated ISPC-accelerated BC7 compression into the Mesa graphics driver; verified both through quality and performance benchmarks.
- Profiled OpenGL workloads on Linux using gdb and trace analysis; built Python/Bash tooling for automated quality evaluation and benchmarking.
- As a parallel cross-team project, developed IDet, a C++17 CPU inference library for text, face, and clothing detection using OpenCV, ONNX Runtime, and OpenMP.
- Optimized tiled inference and threading to **p99 below 10 ms per detector on FHD frames** on Kunpeng 920B.
- Built a Python pipeline for dataset preparation, YOLO training, evaluation, and ONNX export over **270k images**.

Additional Experience

Resident and Ambassador — NEIMARK IT-Academy • Sep 2022 — Sep 2024

Delivered team software projects, mentored hackathon participants, and contributed to a winning AI ethics project. Program Page | Certificates

Private Tutor — Mathematics and Computer Science • Sep 2022 — Present

Prepared 40+ students for mathematics and computer science exams; taught algorithms and adapted learning plans to individual needs.

Projects

IDet — CPU ROI detection library | GitHub

CPU-only library for ROI detection. Stack: C++17, ONNX Runtime, OpenCV, OpenMP, Meson.

YOLO Training and Deployment Pipeline | GitHub

End-to-end pipeline for dataset preparation, training, evaluation and ONNX deployment. Stack: Python, Ultralytics YOLO, ONNX Runtime, Bash.

iVoice — Local Audio and Text Command Processing | GitHub

Local-first pipeline for speech recognition and command normalization. Stack: Python, faster-whisper, Transformers, FastAPI.

Details

Nizhny Novgorod, Russia

Contacts

+7 930 810 67 22

daniil.d.parshin@gmail.com

Links

GitHub | LeetCode | Telegram

Technical Skills

Programming

C++, C, Python, Go (Golang), Groovy, Bash; algorithms & data structures

Build, CI/CD & Testing

CMake, Meson, Jenkins, Docker, Git, GTest, Robot Framework

Tools & Collaboration

Jira, Confluence, Agile (Scrum / Kanban)

Systems & Performance

Linux, gdb, perf, VTune, NUMA, profiling, benchmarking, trace analysis, OpenMP, multithreading

Graphics

Mesa, OpenGL, BCn

Computer Vision & ML

OpenCV, ONNX Runtime, YOLO, DBNet, SCRFD, model training & export, CPU / GPU inference

Languages

Russian — Native

English — B1

Spanish — A1

🎖 Awards and Training

Alice & Yandex Smart Devices — Yandex × HSE × MIPT | Apr 2026

Completed a 72-hour intensive on smart-device architecture, embedded systems, and resource-constrained ML deployment; developed and defended a team R&D project. Certificate

Hack Lab — Sber × School 21 | Nov 2025

Won 1st place with the team in an HPC hackathon covering parallelization, OpenMP, hardware-level optimization, random number generation, and data integrity. Certificate

YADRO Winter School — Programming for RISC-V | Feb 2025

Completed intensive training in RISC-V programming, computer architecture, and low-level systems with practical project work. Certificate

Samsung IT-School — Mobile Development | Sep 2019 — Jun 2020

Studied Android, Java, OOP, algorithms, and data structures. Built a Firebase messenger and won 1st place in the regional round. Certificate

🎓 Education

HSE University — Nizhny Novgorod, Russia | Sep 2026 — Present

Master's Degree in Applied Mathematics and Information Science. Program: Data Mining

HSE University — Nizhny Novgorod, Russia | Sep 2022 — Jun 2026

Bachelor's Degree in Applied Mathematics and Information Science | Thesis